

No. of Experiment:

Name of the Experiment: “Determination of Focal Length and Power of a Convex Lens by Displacement Method”.

Theory:

The distance between the optical centre to the principal focus of a lens is called the focal length of that lens and it is expressed by f . If the focal length of a lens is expressed in metre and sign of inverse value is changed then the power of the lens is obtained in diopter.

If u and v are respectively the object distance and image distance respectively then for real image, we get –

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$f = \frac{uv}{u + v}$$

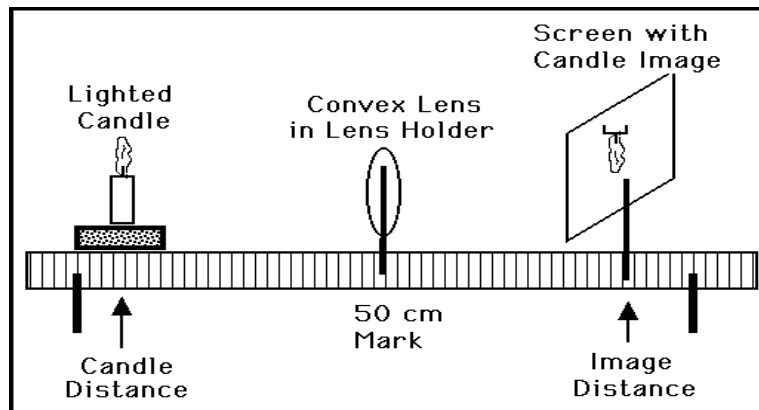


Fig. 1: For Focal length determination.

The power of the divergence or convergence of light ray by a lens is called its power. As the focal length of the lens is less, the less is the distance within which parallel rays become converge or diverge. That means Power of the lens is more. That is why, the inverse of focal length of a lens is called its power. So, if focal length of a lens is known power of that lens can be determined by the following equation.

Power of a lens,
$$P = \frac{1}{f(m)} \text{ Dioptre}(D)$$

Since the lens is a convex lens, so its power will be positive.

List of Apparatus:

1. Optical bench
2. Experimental convex lens
3. Candle as object
4. Observing Screen
5. Index rod

Procedure:

1. A convex lens is taken and it is placed on a rod placed on the optical bench.
2. In front of the lens and on the optical bench a candle is placed on another rod.
3. On the other side of the lens a screen placed on a rod is placed on the optical bench in such a way as the real image of the candle light become incident on it avoiding the parallax error.
4. From the scale on the optical bench position of the lens, object-candle and image screen are taken and apparent values of u and v are found out.
5. A few more apparent values of u and v are found out following the above procedures.
6. From the values of u and v , the focal length, f are found out. The inverse of this value of focal length (in metre) gives the power of the lens.

Observation and Manipulation:

Table 1: For the determination of the Focal length & Power of the Convex lens.

Object side	No. of Obs.	Positions			Distance between Lens & Object, $ X - L = u$ (cm)	Distance between Lens & Image $ X - I = v$ (cm)	Focal length, f (cm)	Average Focal length, f (cm)	Power, $P = \frac{1}{f} D$
		Lens position, X (cm)	Object position, L (cm)	Image position, I (cm)					
1st side	1.								
	2.								
	3.								
	4.								
	5.								
	6.								

Calculations:

Results:

1. The focal Length of the given lens, $f =$ (cm)
2. Its Power, $P =$ (D)

Precautions:

1. Object-candle and image-screen should be in the same straight line the principal axis of lens.
2. There should not be any parallax the tips of the object and image.
3. Readings should be accurate.
4. Index error should be found out.

Discussion:

1. If the index errors are not determined then object-distance and image-distance will not be correct. So, the experiment result will contain error.
2. If the experimental readings are taken accurately then the result will not be correct.